

Sam BOUCHAUD

Mechanical & Robotics Engineer

Engineering degree and MSc in Robotics (2026), seeking a first position in mechanical engineering and robotics around Montpellier. **Portfolio:** `sboqu.github.io`

EXPERIENCE

2026 — **Final-Year Internship (5 months)**

Research Intern, Robotics Lab, University of Padua (Italy)

- Co-built a **Simscape digital twin** of a cable-driven parallel robot and generated a **2.7M-sample** dataset.
- Benchmarked **6 deep learning models** and fine-tuned an **S5 model** on the real robot.
- Pose error of **0.44 cm** and **0.64°**, validated with **Vicon**.

2026 ↑ 2025 — **Third-Year Engineering Project**

Dynamic digital twin of a 7-axis industrial robot

- Built the **Pinocchio** model from the **SolidWorks** CAD (**URDF**).
- Computed joint torques by **RNEA** on filtered **RoboDK** trajectories to size the motors.

2025 — **Second-Year Internship**

Assistant Engineer, VINCI Energies (TG Concept)

- Redesigned a **baggage sorting system**: 3D CAD, client and supplier coordination.
- Developed a **Python** tool generating supplier files from ERP data and 3D/STEP files.
- Saved **30 hours** and up to **€3,500** per project.

2025 ↑ 2023 — **Reservist, 92nd Infantry Regiment**

French Army: discipline, stress management, team cohesion

PROJECTS

Two-Stage Cycloidal Gearbox *(Interactive 3D online)*

- Designed a **1:100** gearbox for an industrial robot shoulder.
- Sized every part: **Hertz** and **EHL** contacts, **ISO 281** bearings, von Mises shafts, Ashby material choice.
- **Differential evolution** optimisation (14 variables, 13 constraints) and 96-part CAD in **3DEXPERIENCE**.

4WS Agricultural Robot Simulator *(Details online)*

- Closed-loop **Scilab** simulator with 4 control laws.

EDUCATION

2026 ↑ 2023 — **Engineering Degree in Advanced Mechanics**

INP SIGMA Clermont, Aubière, major in Mechatronics and Robotics

2026 ↑ 2025 — **MSc in Artificial Perception and Robotics**

Université Clermont Auvergne (Double Degree)

2023 ↑ 2022 — **CPGE ATS, Industrial Engineering**

Lycée Philippe de Girard, Avignon

CONTACT

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Montpellier area, France

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Portfolio
`sboqu.github.io`

SKILLS

CAD
3DEXPERIENCE, SolidWorks, Inventor

Mechanical Design
Hertz contact, Bearing selection (ISO 281), Von Mises, Material selection (Ashby)

Simulation
MATLAB, Simulink, Simscape, Scilab, Pinocchio, RoboDK

FEA
ANSYS, Inspire

Robotics & Control
Robot modelling, Non-linear Control, State Observers, Digital Twins

Programming
Python, C (Arduino), SQL

Machine Learning
PyTorch, Scikit-learn

Workshop
Milling, Turning, CNC

LANGUAGES

French: Native
English: Proficient (TOEIC 935)
Italian: Intermediate

INTERESTS

Swimming, Team running, Geopolitics